Data Science Community for Health and Care Newsletter September 2026

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Welcome to the latest newsletter from the Data Science Community for Health and Care, brought to you by the NHS England Data Science Professional Development Functional Team.

The newsletter team are always happy to receive constructive feedback, and we invite you to send us any contributions you may have.

If you cannot access something of interest to you, please reach out.

Thanks for reading! – newsletter team

October Analyst X Data Science Huddle

Tuesday 27th October 2026, 13:30 - 14:15, Online

The Data Science Community for Health and Care have organised the next Analyst X Data Science Huddle for October. We will hear from one project followed by 2 practical examples of how it has been used in the NHS; see the abstract below.

- **Process Mining in the NHS**

A high-level introduction to process mining: what it is, how it works, and how it can be used to understand the way NHS services operate.

This will be followed by 2 practical examples of how it has been used in the NHS.

1. Outpatient Pathway Protocols

This example will demonstrate how process mining is applied to outpatient pathway protocols. By comparing the intended pathway with observed activity, we can identify where follow-up appointments are concentrated, understand how patients move through different pathway variants, and highlight opportunities for providers to focus improvement efforts.

The aim is not to prescribe a single solution, or a single analytical method, but to provide providers with clearer evidence about where opportunities may exist to reduce avoidable follow-ups, improve pathway efficiency, and release capacity as part of their outpatient planning.

2. Understanding Emergency Respiratory Patient Flow - Applying process mining and collaborative analytics to uncover patient flow patterns and identify improvement opportunities

This is presented as an example at Nottingham University Hospitals (NUH) of executing a 'project focused' deep dive to gain insights from our rich set of disparate data.

Buit upon a robust, version-controlled data pipeline, the analysis mapped patient journeys across the respiratory pathway. This included mappings of patient movements, pathway timings, length of stay, occupancy, and daily activity patterns to provide a detailed evidence base for understanding service performance and flow.

A key principle of the project has been close collaboration with clinicians and operational teams. Recognising that data alone does not provide sufficient understanding of clinical pathways, the analysis was delivered through a series of iterative sprints. Regular review sessions enabled findings to be discussed, assumptions challenged, and priorities refined. This approach developed a shared understanding of both the service and the data, while allowing the analytical focus to adapt as new insights emerged.

This helped identify opportunities for service improvement, challenge anecdotal perceptions, as well as providing a foundation that could be built upon for future projects and more advanced predictive modelling.

November Analyst X Data Science Huddle

Tuesday 10th November 2026, 13:30 - 14:30, Online

The Data Science Community for Health and Care have organised the next Analyst X Data Science Huddle for November. We will hear from two projects:

Using classification and regression trees (CART) to investigate unknown ethnicity records in NHS emergency care data Presented by Grace Dean

Time: 13:30 – 14:00

High-quality ethnicity data is necessary for tackling health inequalities. This study analysed 24.2 million Emergency Care Data Set (ECDS) records from 2023/24 to (a) investigate variation in unknown ethnicity records, (b) explore patterns of missingness between ethnicity and other variables, and (c) identify the variables most important in the recording of an unknown ethnicity.

Variation in unknown ethnicity recording was identified by age, organisation type, and attendance characteristics. Unknown ethnicity was associated with increased missingness across 62 other ECDS variables. Provider site code emerged as the strongest predictor of unknown ethnicity recording, with a small number of acute trusts and independent sector organisations contributing disproportionately.

Recommendations for future work focus on improving data collection processes for individual organisations, which would provide direct benefits for NHS England and NHS providers, and is necessary for tackling health inequalities.

Here's a link to the paper in BMJ Open, and on the NHS England Data Science page.

Can Routine Pathway Data Identify ED Attendance Risk While Patients Wait? Presented by Kayleigh Banyard-Haydock

Time: 14:00 – 14:30

The project investigates whether routinely collected healthcare data can be used to identify RTT elective waiting-list pathways at increased risk of emergency department attendance while awaiting treatment.

It focuses on the development and evaluation of a prediction model that produces an individualised probability risk score using information available at the beginning of the waiting-list pathway. The model aims to support risk stratification and help identify patients who may benefit from earlier review or additional support, while providing a foundation for further model development. The work also considers how predicted risk could support service planning and prioritisation across these pathways.

Fancy an interesting read?

Take a look at the Why mental metaphors do not help us understand chatbot mistakes article from Springer.

Factual mistakes made by ChatGPT are frequently communicated using metaphors describing mistakes made by human minds, for example, hallucination.

The article discusses this notion and demonstrates that the two most common metaphors dont't manage to provide us with a better understanding of factual chatbot mistakes. They argue that ChatGPT – and other chatbots such as Google's Gemini by extension – do not hallucinate when they generate factually incorrect outputs, not even in a metaphorical sense. The article draws on research on hallucination in philosophy and the empirical mind sciences.

Datacise Open Learning Courses

Datacise Open Learning provide training for data science in research. There are a variety of courses coming up! For more information, including how to register, follow the course link.

Data quality aspects of administrative data (London)

6th October 2026, 09:00 - 17:00, UK Research and Innovation (UKRI), London, SW1W 9SP

This course will provide an introduction to data quality, and how it can affect all aspects of working with administrative data. The course will cover data quality dimensions which include technical and social aspects.

Introduction to Government, Policy & Impact

11th November 2026, 09:30 - 15:15, Live Online

This course explores different experiences of successfully engaging with policy makers, so that research evidence has a greater impact and influence. The course aims to be engaging, informative, interactive, and practical, with opportunities to share experiences from your research.

Ethical AI Coding with Synthetic Administrative Data

20th November 2026, 09:30 - 16:00, UK Research & Innovation, London SW1W 9SP and Online

This course explores the ethical use of local AI models for developing code to analyse synthetic administrative data, combining practical skills training with examples of best practice.

Good Practice for R Users When Working in the ONS SRS

25th November 2026, 14:00 - 15:00, Online

A practical workshop on writing efficient, reproducible and robust R code when working with large administrative datasets in the ONS Secure Research Service. Drawing on real-world SRS experience, the session will cover analytical workflow design, efficient database queries, memory management, safe joins, validation and reproducibility.

ADR UK Determining Healthy Households

Self-paced, Online

A self-paced course on how to create Household Units of Analysis (HUAs) from line-level pseudonymised GP patient data.

Events

Lots of exciting things coming up! See the full calendar here, and a small selection below.

AI in the NHS 2026

Thursday 15 October 2026, 10.00–17.00, Online

2026 is set to be a pivotal year for AI in the NHS, with important national strategies and regulatory reforms expected that will impact what kinds of technologies get used and how the health service adopts and oversees them.

But turning the vision of an AI-enabled health system into reality won't be straightforward. Our third annual AI event will bring together stakeholders from across the NHS, government, industry, academia and the charity sector to consider how the NHS can responsibly, safely and effectively implement AI at scale.

This full day event will explore how to:

- translate national strategy into real-world implementation
- evaluate and scale AI tools rapidly, safely and effectively
- build trust among clinicians, patients and the public
- support the workforce through technological transformation
- learn from leading international health systems already using AI at scale.

Webinar: Achieving a low-carbon, climate resilient NHS

Thursday 5th November, 12:00 - 13:15, Online

The health of people in the UK is being actively impacted by climate change, and this is only set to worsen over the coming years. The Lancet Countdown shows that Europe is facing the fastest changing climate globally. Healthcare contributes 4-6% of global greenhouse gas emissions, more than the aviation industry.

In response to this, the NHS made the world's first legally binding commitment to becoming a net zero health system – by 2040 for emissions it controls, and 2045 for emissions it influences. Action has been underway since 2020, but more needs to be done to achieve this goal, requiring transformation at scale to sustainable models of care, with supportive leadership, policy and innovation. At the same time, the NHS must move to protect health and maintain service delivery during heatwaves, floods and storms.

This webinar brings together people from policy, research, clinical and innovation backgrounds to share what we can learn from work building a low-carbon, climate resilient NHS.

Common challenges and policy responses: international lessons for the NHS

Thursday 26th November, 10:00 - 17:00, Online

The NHS in England is under huge pressure. Health needs are growing as more people live longer and with major health conditions. Staff are under significant strain. Funding increases are modest by historic standards and productivity remains below pre-pandemic levels. Meanwhile the outlook for the UK economy is weak and the public finances are stretched, leading some commentators to question the NHS's long-term sustainability. AI and new technology offer opportunities for health and care, but also challenges for policymakers seeking to harness and regulate them.

The UK is not alone here. Many countries are facing similar long-term trends and seeing their health systems coming under sustained pressure. In response, policymakers across Europe and elsewhere are implementing a range of reforms to their health systems to improve and redesign care, stabilize services, drive greater efficiency and control costs.

This summit will be an opportunity to hear about the major challenges facing health systems internationally, learn about policy responses in other countries, and discuss what lessons they hold for policy and practice in England. Senior policymakers and leaders in the NHS will join to reflect on how services in England need to adapt.

See more future events on the calendar

Know of any events we should feature next month? Let us know by clicking the "Contribute" button, or here.

Check out our collection of training resources in the Resources Section! Can you spot something missing? Contact us!

Need a Quick Break?

How many tries will it take you?

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